Team 11: Team Devops

Team	Team Devops
Project	Digital Knowledge Platform for CSEDU (with AI)
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Core features to consider

1. Add AI response confidence scoring. The RAG service returns answers with citations, but users have no way to judge reliability. Display a confidence indicator (high, medium, low) alongside each response, based on retrieval similarity scores, so users know when to verify manually.
2. Build a query analytics dashboard for admins. The SRS logs AI usage (FR-AI-015), but there is no interface for analyzing query patterns. Show most-asked questions, failed queries, and popular content so admins can identify gaps in the knowledge base.
3. Implement streaming responses for long summarizations. Document summarization can take 5-10 seconds (NFR-TXX-041). Streaming the output in real-time instead of showing a loading spinner would significantly improve perceived performance.
4. Add AI-generated metadata tagging on document upload. Researchers currently enter keywords manually. Use the RAG pipeline to auto-extract keywords and subject tags from uploaded text, then let users review and edit before saving. This speeds up metadata entry and improves search quality.

What's missing

- No policy on storing AI-generated summaries. The system generates summaries dynamically (FR-AI-003), but never specifies whether they are cached permanently, cached temporarily, or regenerated on every request. This affects storage costs and user expectations if the model is upgraded.
- Missing escalation path for AI errors. If the AI hallucinates or provides incorrect information, there is no documented process for flagging, correcting, or preventing recurrence. A "report inaccuracy" button that logs issues for review would be a reasonable starting point.
- SSO fallback logic is incomplete. The SDD mentions "local JWT fallback if SSO unavailable," but does not specify timeout thresholds or what happens to users already authenticated via SSO when the identity provider goes down.
- No AI response audit trail specification. The RAG service logs which model was used, but does not specify whether the prompt, retrieved document chunks, or model parameters are recorded. For AI governance and debugging, this level of detail matters.
- **No Git Repository**
- No API documentation is provided in the document.

Weaknesses of SRS

The AI hallucination handling in Section 4.2.5 is underspecified to the point of being hand-wavy. FR-AI-005 requires "answers with source references" and FR-AI-008 says "low-confidence responses should be flagged," but the SRS never defines what triggers a low-confidence flag. Is it a model confidence threshold? A retrieval similarity cutoff? The acceptance criterion ">=95% responses include citations" says what to count but not how the system prevents hallucinations in the first place. For an academic platform where accuracy matters, this gap is serious.

Weaknesses of SDD

The SDD mentions "HNSW index (m=16, ef_construction=64)" for vector search, but does not explain why these parameters were chosen for the expected data scale. There is no benchmarking data, no growth plan for when the index exceeds 100K documents, and no monitoring strategy for query latency degradation. The document acknowledges a potential migration to Weaviate at 1M+ documents, but says nothing about the 100K-to-1M range where performance will actually degrade.